Lecture no. 14 :-

Random Variables, Basics and Rationale

Events vs Variables

- Assign outcomes of experiments to variables
- But why?
- Example: 6-side dice rolled

$$\overset{\downarrow}{N} \in \overset{\text{type}}{\mathbb{N}}, \quad \overbrace{1 \leq N \leq 6}^{\text{constraint}}$$

Maths —— Constant —— Variable

Programming —— Constant —— Varible

"Never changes" — "Assigned once never changes afterwards" — "Can always change"

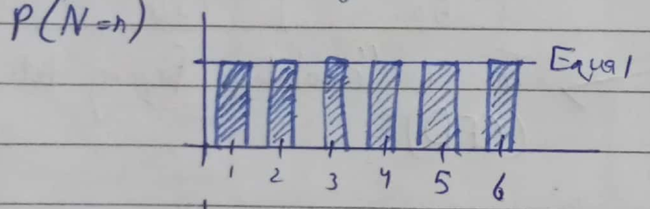
Random variable (RVs) In between Constand & Variable in Programming

- A RVs can take on a value from any set
  Each value has a probability associated with it.

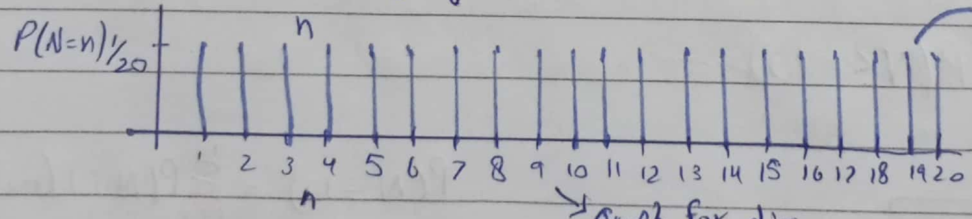
e.g

N	1	2	3	4	5	6
$P(N=n)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

- Other examples:
  - Pick any person and measure their height
  - Pick any character and convert it to ASCII.



- Similar to the histogram we saw earlier...



This is called a "distribution".

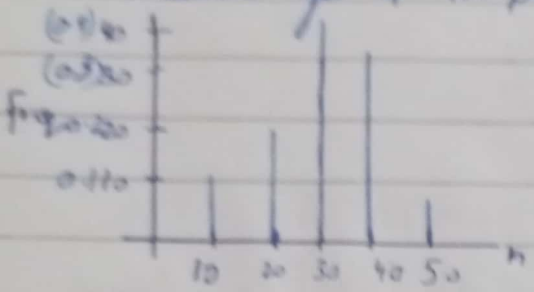
Graph for dice with 20 sides

The divide of probability into the ~~[illegible]~~ parts for each random variable is called distribution.

N is a "discrete random varible".

(Side note: Another example)

Measure ages of 100 people

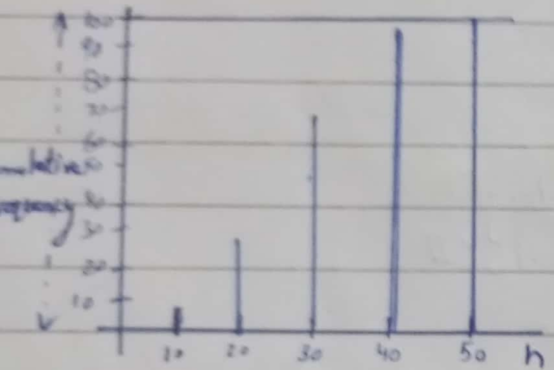


Frequency Table

Weight	Frequency
10	5
20	20
30	40
40	30
50	5

This is a "probability frequency distribution" also known as (PFD)

(frequentist view)

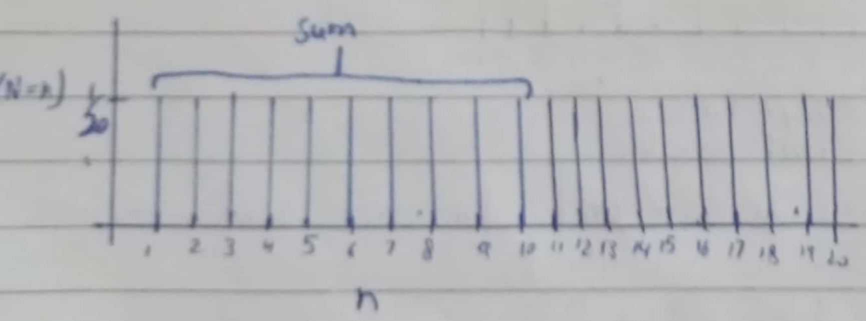


Weight	Frequency
10	5
20	25
30	65
40	95
50	100

This is "cumulative frequency distribution" (CFD)

Back to distribution:

$P(N <= 10)$



$P(N \leq 10) = \sum_{i=1}^{10} P(N=i)$ (mutually exclusive)

"$P(N \leq 20)$ must equal 1"

$P(N<1)$ " " 0